Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 119506

Subject: Environmental Engineering-I

Time: 03 Hours
Full Marks: 70

Instructions:-
(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The distribution mains are designed for
- (i) maximum daily demand
- (ii) maximum hourly demand
- (iii) average daily demand
- (iv) maximum hourly demand on maximum day

(b) Orthotolidine test is used for determination of
- (i) dissolved oxygen
- (ii) residual chlorine
- (iii) biochemical oxygen demand
- (iv) dose of coagulant.

(c) Period of cleaning of slow sand filters is about
- (i) 24 to 48 hours
- (ii) 10 to 12 days
- (iii) 2 to 3 months
- (iv) 1 to 2 year

(d) Which of the following air pollution control device has maximum efficiency?
- (i) Spray tower
- (ii) Wet cyclonic scrubber
- (iii) Dynamic precipitator
- (iv) Electrostatic precipitator

(e) The maximum permissible limit of fluoride in drinking water is
- (i) 0.1 mg/l
- (ii) 1.5 mg/l
- (iii) 5 mg/l
- (iv) 10 mg/l

(f) The dissolved oxygen level in natural unpolluted water at normal temperature is found to be of the order of
- (i) 1 mg/l
- (ii) 10 mg/l
- (iii) 100 mg/l
- (iv) 1000 mg/l

(g) As compared to geometric increase method of forecasting population, arithmetical increase method gives
- (i) lesser value
- (ii) higher value
- (iii) same value
- (iv) accurate value

(h) The pipe which is installed in the house drainage to preserve the water seal of traps is called
- (i) vent pipe
- (ii) Anti-siphonage pipe
- (iii) waste pipe
- (iv) soil pipe

(i) The velocity of flow of water in sedimentation tank is about
- (i) 5 to 10 cm/s
- (ii) 15 to 30 cm/min
- (iii) 15 to 30 cm/s
- (iv) 15 to 30 cm/hr

(j) What type of plume behaviour occurs during super-adiabatic (unstable) atmospheric conditions?
- (i) Looping plume
- (ii) Fanning plume
- (iii) Coning plume
- (iv) Lofting plume

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Q.2 (a) What are the steps involved in the primary, secondary and tertiary treatment of water? Explain with a neat flowchart. [7]

(b) Explain in detailed the procedure for determination of MPN index of a water sample. [7]

Q.3 (a) What are different methods of population forecasting? Compute the population of the year 2000 and 2018 for a city whose population in the year 1930 was 25,000 and in the year 1970 was 47,000. Make use of geometric increase method. [7]

(b) What is meant by per capita demand and design period? Discuss various factors which affect per capita consumption of water. [7]

Q.4 (a) Calculate the alkalinity, total hardness, carbonate hardness, and non-carbonate hardness for the following water in mg/l as $CaCO_3$. [7]

Cation	mg/l	Anions	mg/l
Ca^{2+}	74	HCO_3^-	126
Mg^{2+}	48	SO_4^{2-}	134
Na^+	14	Cl^-	92
K^+	31	pH	7.0

(b) Explain the various methods of water softening. Enumerate their relative advantages and disadvantages. [7]

Q.5 (a) Design rapid sand filters beds for treating water required for a population of 1,50,000 people with an average rate of demand of 150 lpcd with the following data: [7]

(i) Filter area of each unit should not be more than 50 m^2
(ii) Amount of water used for back-washing is 4% of treated water per day
(iii) The time required for back washing may be neglected
(iv) One unit must be kept as stand by
Assume any other data required.

(b) For disinfecting water supply, it is required to treat 500,000litres of daily supply with 0.5ppm of chlorine. If the disinfectant is available in the form of bleaching powder containing 30% of available chlorine, calculate the amount of bleaching powder required to treat the daily flow of water. [7]

Q.6 (a) What is the difference between coagulation, flocculation, and clarification in water treatment process? List any three coagulants used in water treatment. [7]

(b) 4 MLD of water per day passes through a sedimentation tank basin which is 6 m wide, 16 m long and 3.5 m deep. (i) Find the detention time for this basin (ii) Determine the average velocity of flow through the basin (iii) Compute the SOR of the basin. https://www.akubihar.com [7]

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Q.7 (a) Describe with neat diagrams various layouts of distribution systems. Mention the applicability of each. [7]

(b) Explain different types of plumbing fixtures and fittings used in modern buildings. [7]

Q.8 (a) Explain air quality standards and various techniques used for air pollution control. [7]

(b) Differentiate between sound and noise. Discuss any five methods of noise control. [7]

Q.9 Write Short notes on ***any two*** of the following: [7 x 2 = 14]

(a) Aeration Methods
(b) EDTA method for determination of Total, Temporary and Permanent Hardness
(c) Flow Mass Curve and Coincident Draft
(d) Pressure Reducing Valves and Break Pressure Tanks

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